

NTIRE 2026 Remote Sensing Infrared Image Super-Resolution Challenge Factsheet: Fusing Progressive Focus and Broad Pixel Activation Transformer for Remote Sensing Infrared Image Super-Resolution

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1. Introduction

This factsheet describes the solution of team **XJRes** (ID: 03) for the NTIRE 2026 Remote Sensing Infrared SR Challenge ($\times 4$). The proposed method HAT_PFT combines two complementary super-resolution models, namely the Progressive Focused Transformer (PFT [1]) and the Hybrid Attention Transformer (HAT [2]), trained from scratch on the provided dataset. Our method scores **54.29** (PSNR + $20 \times$ SSIM) on the official leaderboard vs. bicubic baseline 50.39.

2. Email Submission Guide

To: normal.kliu@gmail.com, yuehaoyang@sjtu.edu.cn, tshzary@gmail.com, zhengchen.cse@gmail.com, jing kai-wang100@gmail.com, g1017325431@gmail.com, Radu.Timofte@uni-wuerzburg.de, yulun100@gmail.com

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Subject: NTIRE 2026 Remote Sensing IR SR – XJRes – 03

Body:

- Team: XJRes
- Leader: Li Pang, pp2373886592@gmail.com
- Members: Heng Zhao, Xinqiao Wu, Deyu Meng, Xiangyong Cao
- CodaBench user: testify
- Code: `git clone https://github.com/TESTIFY-123/NTIRE2026-Infrared.git`
Model: `gdown --id 1SUh7cuFPA3b-I9WI1mkH-qESEppTrxY`

f) Results: `gdown --id 1yszWT-uKl82YbSOjLnJw7CPIrvTUvwDE`

3. Code Submission

- `git clone https://github.com/TESTIFY-123/NTIRE2026-Infrared.git`
- Model: `models/team03_HAT_PFT/io.py`
- Checkpoint: `model_zoo/team03_HAT_PFT/HAT_PFT.pth` (link in `team03_HAT_PFT.txt`)
- `python test.py --model_id 3 --test_dir [path] --save_dir [path]`
- `python eval.py --output_folder [path] --target_folder [path] --gpu_ids 0`

4. Team Details

- Team:** XJRes **ID:** 03
- Leader:** Li Pang
- Email:** pp2373886592@gmail.com
- Affiliation:** Xi'an Jiaotong University
- CodaBench:** testify
- Best score:** 54.29 (PSNR + $20 \times$ SSIM)
- External data:** None
- Pretrained models:** None (scratch)

5. Method

5.1. Overview

The proposed method combines two complementary super-resolution models, namely the Progressive Focused Transformer (PFT [1]) and the Hybrid Attention Transformer (HAT [2]). Given a low-resolution input image I_{LR} , the

image is processed by four parallel branches. The first two branches adopt the PFT [1] architecture, while the last two branches employ the HAT [2] architecture to generate super-resolved results. The outputs from these four branches are then fused through a weighted averaging strategy to obtain the final high-resolution image. This design enables the framework to simultaneously exploit the progressive attention focusing capability of PFT [1] and the hybrid attention representation ability of HAT [2], thereby enhancing the reconstruction performance.

5.2. Architecture

5.2.1. Progressive Focused Transformer Branch

The first two branches adopt the Progressive Focused Transformer (PFT [1]) architecture for image super-resolution. Note that these two PFT [1] branches share the same network architecture but are trained independently with different objective functions to capture complementary features. As illustrated in Fig. 1, the network progressively extracts image features and reconstructs the high-resolution output.

- **Feature Encoding:** Given the low-resolution input image I_{LR} , a convolutional layer is first applied to transform the input into a high-dimensional feature representation F_0 . This stage extracts basic structural and texture information for subsequent feature learning.
- **Progressive Attention Representation Learning:** The core of this branch consists of multiple Progressive Focused Attention (PFA) blocks. Each block contains several Progressive Focused Attention Layers (PFAL), where attention maps are propagated across layers to progressively refine the focus on important tokens. Through attention inheritance and sparse selection, the PFA mechanism enhances highly relevant features while suppressing irrelevant tokens, enabling efficient feature aggregation.
- **Image Reconstruction:** After deep feature extraction, the refined features are passed through convolutional layers and an upsampling module to generate a super-resolved image I_{SR}^{PFT} .

5.2.2. Hybrid Attention Transformer Branch

The remaining two branches are based on the Hybrid Attention Transformer (HAT [2]). Similarly, these two HAT branches share the same network architecture but are trained independently with different objective functions. The model integrates both channel attention and window-based self-attention to improve feature representation.

- **Shallow Feature Extraction:** The input image I_{LR} is first processed by a 3×3 convolutional layer to obtain shallow features F_0 , mapping the image into a higher-dimensional feature space.
- **Deep Feature Extraction:** This stage contains several Residual Hybrid Attention Groups (RHAG), each consisting of multiple Hybrid Attention Blocks (HAB) and an

Overlapping Cross-Attention Block (OCAB). The HAB combines window-based self-attention and channel attention to capture both local textures and global dependencies, while the OCAB enhances cross-window interactions for more effective feature aggregation.

- **Image Reconstruction:** Finally, a convolution layer followed by a pixel-shuffle [3] operation upsamples the deep features to produce the super-resolved output I_{SR}^{HAT} .

5.2.3. Weighted Fusion

After obtaining the outputs from the four branches, the final super-resolved image is generated by a weighted fusion strategy combined with a self-ensemble technique during inference, which aggregates predictions from geometrically transformed inputs. Let $I_{SR}^{(i)}$ denote the super-resolved output of the i -th branch. The final result is computed as the weighted average of the four outputs:

$$I_{SR} = 0.25 I_{SR}^{(1)} + 0.25 I_{SR}^{(2)} + 0.25 I_{SR}^{(3)} + 0.25 I_{SR}^{(4)}. \quad (1)$$

where the first two branches correspond to the PFT model and the remaining two branches correspond to the HAT model. Since all weights are equal, the fusion operation is equivalent to averaging the outputs of the four branches.

5.3. Training

- **Data:** 1,019 official HR/LR pairs, no extra data
- **Loss:** PFT branches: RBSFormer [4] & L_1 ; HAT branches: RBSFormer [4] & CharbonnierL1 + SSIM
- **Training:** lr= 2×10^{-4} (Cosine Annealing), 200ep, bs=1, FP32
- **Optim:** Adam [5]
- **HW:** RTX 4090 GPUs (PyTorch)

5.4. Inference

LR (3-ch PNG) $\rightarrow$ /255 $\rightarrow$ avg3-ch $\rightarrow$ model $\rightarrow$ $\times 255 \rightarrow$ 1-ch PNG. 8-way TTA (rotation + flip).

5.5. Results

Method	PSNR	Score
Bicubic	—	50.39
HAT_PFT (ours)	35.87	54.29

$$\text{Score} = \text{PSNR} + 20 \times \text{SSIM}.$$

5.6. Complexity

Parameters: $\sim 115\text{M}$. Runtime: ~ 285 min / 222 images on RTX 4090. GPU: Yes.

6. Other Details

None

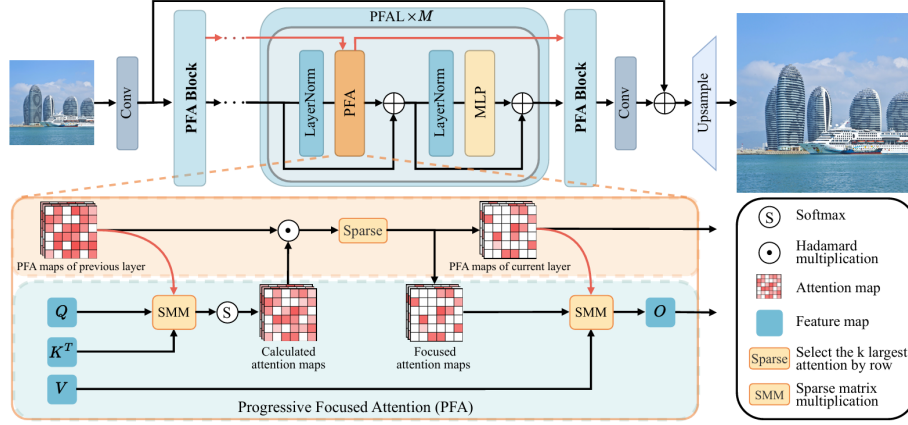


Figure 1. Overall architecture of the Progressive Focused Transformer (PFT).

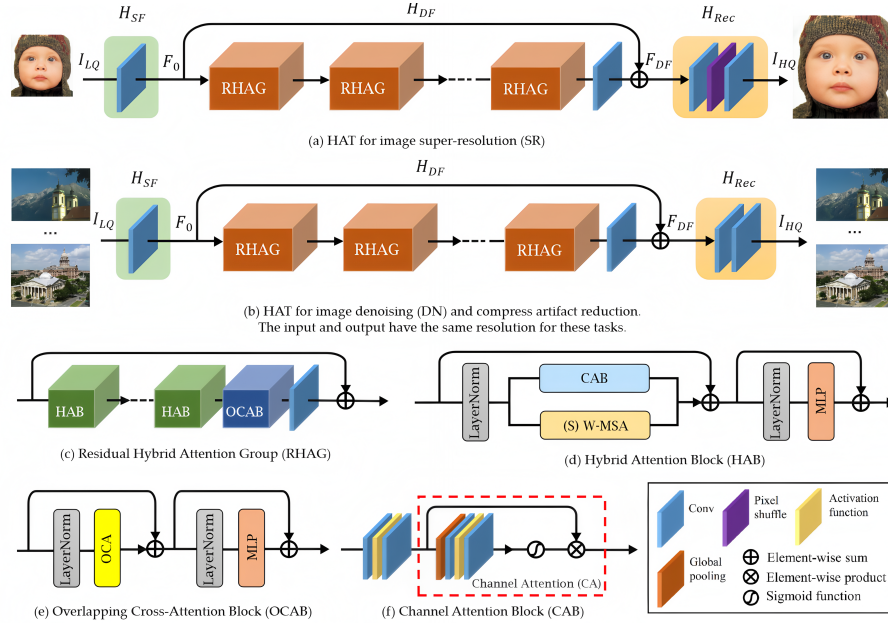


Figure 2. The overall architecture of HAT and the structure of RHAG and HAB.

References

- [1] Long, Wei and Zhou, Xingyu and Zhang, Leheng and Gu, Shuhang Progressive Focused Transformer for Single Image Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2279–2288, 2025. 1, 2
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- [3] Wenzhe Shi, Jose Caballero, Ferenc Huszár, and others. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1874–1883, 2016. 2
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- [5] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. 2